



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.C.A. (CBCS) Degree Examinations

January 2024

22MCA103 – COMPUTER ORGANIZATION AND ARCHITECTURE

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit

Unit – I		Marks	BT*	CO*	PO*
1. a)	Realize XOR and XNOR gates using NAND gates.	12	L*3	1	2
b)	Convert the following:				
	i) (632)8 into decimal.				
	ii) 5C16 into binary.				
	iii) 558 into Hexadecimal.				
	iv) 101010 into Octal.	08	L3	1	1
2. a)	Realize the basic gates using NAND gate and NOR gate.	09	L3	1	2
b)	Show that $AB+A'C+BC = AB+A'C$ using Boolean laws.	06	L3	1	1
c)	Simplify the expression using Boolean Laws and realize the same using NOR gates: $Y = AB+AC+BC'$.	05	L3	1	2
Unit – II					
3. a)	Simplify the following expression using K maps. Also show the intermediate steps:				
	i) $F(A,B,C,D) = m(5,7,8,10,11,12,13,14,15)$				
	ii) Minimize the following function in POS form				
	$F(A,B,C,D) = m(0,1,2,3,4,5) + d(10,11,12,13,14,15)$	10	L3	2	3
b)	What is the use of Parity generator and parity checker? Assume a 3-bit message is sent with even parity, realize a circuit for the same with suitable truth table and circuit at the sender and receiver.	10	L2	2	2
4. a)	Simplify the following expression using Tabulation method:				
	$F(A,B,C,D) = m(0,2,3,6,7,8,10,12,13)$	12	L3	2	3
b)	Implement a full subtractor using 3:8 decoder.	08	L2	2	2
Unit – III					
5. a)	Using a block diagram, which shows the register configuration, perform sequential circuit binary multiplication of multiplicand=1101 and multiplier=1011	08	L3	3	2
b)	Convert the following operations on 5 bit signed numbers using 2's complement representation system. State whether overflow occurs or not in each case.				
	i) $(-10) + (-13)$ ii) $(-10) - (+4)$ iii) $(-3) + (-8)$ iv) $(-10) - (+7)$	08	L2	3	2
c)	Explain and realize the half Subtractor circuit.	04	L1	3	1
6. a)	Given Booth's algorithm to multiply two binary numbers, explain the working of the algorithm with an example.	08	L3	3	2
b)	Explain normalization, excess - exponent and special values with respect to IEEE floating point representation.	08	L2	3	2
c)	Explain the design of a 4-bit carry look ahead adder.	04	L1	3	1

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Unit – IV

a)	What is addressing mode? Explain the different addressing modes with appropriate examples.	10	L2	4	1
b)	Explain the concept of subroutine nesting and processor stack with suitable ALP examples	10	L2	4	2
8. a)	Draw and explain the connection between memory and processor via the respective registers.	12	L2	4	1
b)	Write an ALP program to copy 'N' numbers from array 'A' to array 'B' using indirect addresses. (Assume A and B are the starting memory location of an array).	08	L2	4	2

Unit – V

9. a)	Describe the different mapping functions in cache.	10	L1	5	1
b)	What is bus arbitration? Explain different approaches to bus arbitration.	10	L1	5	1
10. a)	Discuss the internal organization of a 2M x 8 asynchronous DRAM chip.	10	L1	5	1
b)	With neat sketches, explain the various methods for handling multiple interrupt requests.	10	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
